

Optimized Deep Learning Approaches for Lung and Colon Cancer Classification using Histopathological Images

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Abstract— This research study explores the application of pre-trained Convolutional Neural Network (CNN) architectures for the classification of lung and colon cancer using histopathological images. Leveraging the LC25000 dataset, which contains 25,000 high-quality images across five categories, the research employs EfficientNetB6, ResNet34, VGG-19, MobileNetV2, and ResNet50 for image classification. Preprocessing techniques, feature extraction, and classification were carried out using CNN-based transfer learning, optimizing computational efficiency and accuracy. The results reveal that MobileNetV2 and ResNet50 outperform other models with an accuracy of 99%, followed by VGG-19 at 98% and ResNet34 at 97%. EfficientNetB6 achieved 93% accuracy, showcasing reasonable performance despite being less computationally intensive. Evaluation metrics, including precision, recall, F1 score, and accuracy, confirmed the models' efficacy in distinguishing cancerous and normal tissues. The findings demonstrate the ability of deep learning as a robust and efficient tool for early and accurate detection of lung and colon cancer, offering significant support for clinical decision-making.

Keywords— Lung Cancer, Colon Cancer, squamous cell carcinoma (SCC) and adenocarcinoma (ADC)

I. INTRODUCTION

Cancer, a preceding cause of morbidity and mortality worldwide, characterizes a complex group of diseases illustrated by the unrestrained growth and spread of abnormal cells. Despite significant advancements in medical science, cancer continues a worldwide health challenge due to its multifaceted etiology and heterogeneity. The worldwide cancer liability is predicted to rise extensively, with an predictable 28.4 million cases anticipated by 2040, marking a 47% raise from 2020 [1]. This increase is attributed to causes such as aging populations, lifestyle changes, and the pervasive influence of environmental carcinogens. The biology of cancer involves intricate processes such as hereditary mutations, epigenetic variations, and dysregulated signaling alleyways, all contributing to the hallmarks of cancer: constant proliferative signaling, circumvention of evolution suppressers, confrontation to cell death, and initiation of metastasis [2]. Understanding these mechanisms has facilitated the enhancement of leveled therapies and immunotherapies, which have transformed the treatment landscape for certain cancer types. However, disparities in outcomes persist, largely driven by limited access to healthcare and the high cost of advanced therapies [3]. Screening and early detection remain critical in reducing cancer-related mortality. Techniques such as

mammography, colonoscopy, and low-dose computed tomography have proven effective in identifying cancers at earlier, more treatable stages [4]. Nonetheless, challenges in widespread implementation highlight the need for innovative diagnostic tools that are both accessible and cost-effective. Emerging research focuses on the integration of artificial intelligence, precision medicine, and nanotechnology in oncology. These innovations offer the potential to revolutionize cancer detection, treatment, and monitoring, paving the way for more personalized and effective care [5]. Lung & colon cancers, preceding causes of cancer-related mortality globally, predominantly manifest as two main histological subtypes: squamous cell carcinoma (SCC) and adenocarcinoma (ADC) [6]. Accurate and timely diagnosis of these subtypes is crucial for determining effective therapeutic strategies. Histopathology, the microscopic examination of tissue samples, remains the gold specification for cancer diagnosis; however, its manual interpretation is often subjective, time-intensive, and prone to variability among pathologists [7]. Advances in artificial intelligence (AI) and computational pathology present promising avenues for automating and standardizing the investigative procedure, particularly through the assessment of histopathology images.

The application of AI in histopathology leverages machine learning and deep learning performances to investigate complex tissue patterns and cellular structures, offering robust tools for distinguishing between SCC and ADC. These techniques have demonstrated the ability to process vast datasets with higher accuracy and efficiency compared to traditional methods [8]. Such advancements not only reduce diagnostic errors but also support pathologists in handling increasing workloads due to rising cancer incidence. Lung and colon cancers share some histopathological features, but their underlying biology and clinical behavior vary significantly. Automated systems that can accurately identify and differentiate these subtypes across different organ systems are critical for improving patient outcomes. Recent studies have utilized CNNs and transfer learning to extract meaningful characteristics from histopathological images, achieving impressive performance in classification tasks [9]. This paper explores automated identification of SCC and ADC in lung and colon cancers by investigating histopathological image data. By evaluating the efficacy of deep learning-based models in classifying these cancer subtypes, this paper aims to provide to the extending body of expertise in computational pathology and support the development of clinically viable diagnostic tools [10].

II. LITERATURE REVIEW

Machine learning algorithms have demonstrated their transformative potential across various sectors, including healthcare, where they enhance diagnostic accuracy and treatment outcomes. These algorithms enable the analysis of extensive datasets, identifying patterns and correlations that traditional methods might overlook. Applications utilizing machine learning have shown remarkable performance in diverse fields, such as healthcare, human-robot interaction, and pattern recognition [11], with significant contributions to the categorization of various cancer types through histopathological image analysis [12, 13]. CNNs have been particularly effective cutting-edge classifying cancer subtypes, including lung and colon cancers. Kuepper et al. [14] developed a class-free colon cancer detection method using 987 histopathological and infrared spectra images, achieving 94% accuracy with Random Forest. Similarly, [15] proposed a Spatially Controlled CNN (SC-CNN) to classify four nucleus types in colon tumors without requiring nuclei segmentation, utilizing 20,000 images and achieving an F-measure of 80.2%. A deep learning approach based on AlexNet to automatically detect tumors in 3,463 frames from colonoscopy footage, achieving a classification accuracy of 91.47% [16].

In the domain of lung cancer, Hatuwal and Bijaya [17] utilized 15,000 images from the LC25000 dataset to train and validate a CNN architecture, achieving 96.11% and 97.2% accuracy, respectively. Similarly, Garg and Garg [18] employed pre-trained CNN architectures such as VGG16 and ResNet50 on over 25,000 histopathology images, reporting accuracy levels between 96% and 100% using advanced augmentation techniques. Nishio et al. [19] introduced a homology-based image analysis (HI) approach, leveraging Betti coefficients to classify lung cancer images, achieving a 99.3% accuracy rate.

Further advancements in cancer detection methods include Akbari et al.'s [20] CNN-based classification approach for colorectal cancer using datasets from the Asu's Mayo Clinic repository, which attained over 90% precision. Bukhari [21] applied ResNet models to classify histopathological images of lung and colorectal adenocarcinoma, achieving sensitivity rates up to 96.77%. For enhanced performance in lung nodule classification, Sethy et al. [22] integrated wavelet features with AlexNet and achieved an accuracy of 99.3%.

Improving classification outcomes via the combination of optimization techniques and ensemble approaches has also been the focus of recent research. Combining logistic regression, support vector machines (SVMs), and random forest classifiers using ensemble approaches achieved a 99.00% accuracy on the LC25000 dataset. Furthermore, XlmNet, an extreme learning machine-based model, was suggested for the classification of lung nodules. By using preprocessing methods such as histogram equalization and PCA, the model achieved a 96.5% accuracy rate [23].

Classification of lung & colon cancers has been shown to constantly improve with the use of feature extraction methods, improved data augmentation, and pre-trained convolutional neural network (CNN) models. The results of these research suggest that automated diagnostic systems that combine machine learning and deep learning might be a game-changer when it comes to cancer diagnosis.

III. PROPOSED METHODOLOGY

The proposed approach involves several stages, including image preprocessing, feature extraction, and classification. A complete explanation of the dataset and each stage of the methodology is provided in the succeeding sections. In this study, pre-trained CNN models, originally trained on the ImageNet dataset [46], are utilized to classify histopathological images. This approach leverages "transfer learning" [47], which uses the learned weights from an existing model to address new tasks. This method reduces computational demands, enhances efficiency, and delivers faster and more accurate results. In this work, CNN architectures such as VGG-19, ResNet34, ResNet50, EfficientNetB6, and MobileNetV2 have been employed to classify lung cancer images into three categories: adenocarcinoma, squamous cell carcinoma, and normal tissue. Similarly, colon cancer images are categorized into two classes: adenocarcinoma and normal. Python-based tools have been used for data preprocessing, and features are extracted using CNN models. These features are then utilized to develop and evaluate the classification models. To further enhance accuracy, regularization techniques have been applied to optimize the learning rate during model training. Multiple CNN models have been compared to identify the most effective approach for classifying the images.

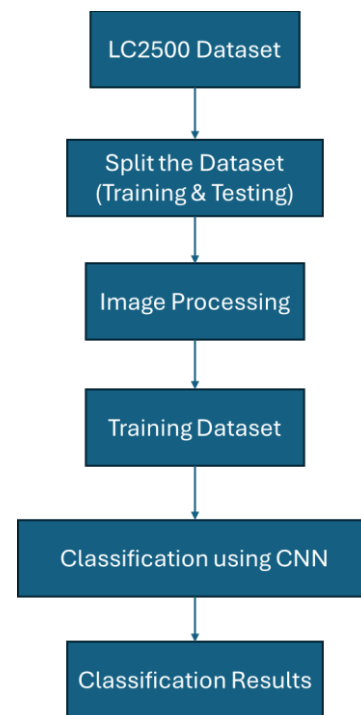


Fig. 1. Proposed Methodology Flow Diagram

IV. DATASET

This research made use of the LC25000 dataset, which contains histology pictures of colon and lung cancers [48]. There are five groups into which the 25,000 histopathological pictures are organized in the dataset. The dimensions of each picture are 768 by 768 pixels, and they are all saved as JPEGs. There were 750 photographs of lung tissue (250 of benign, 250 of adenocarcinomas, and 250 of squamous cell carcinomas) and 500 images of colon tissue (250 of benign, and 250 of adenocarcinomas) in the original sample, which complies with HIPAA validation and

compliance requirements. The dataset was enhanced using the Augmentor program, which allowed for the expansion of the original collection of 1,250 photos to a grand total of 25,000. Training data makes up 80% of the LC25000 dataset, while validation and test data make up the remaining 20%. Figure 2 showcases a representative picture from each of the five categories included in the LC25000 dataset.

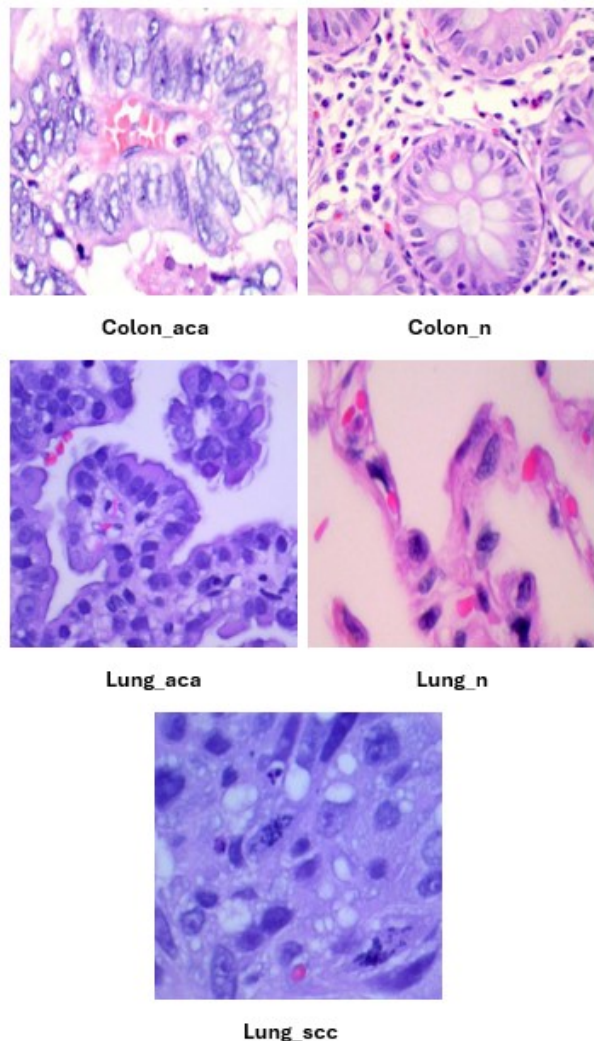


Fig. 2. Sample images of lung & colon from the LC25000 dataset

These results affirm that the MPIF model is well-suited for applications requiring detailed, high-fidelity 3D human models, such as virtual reality, digital fashion, and interactive entertainment. The model's strong performance, particularly in terms of detail resolution and generalization ability, highlights its potential to set new standards in the field of 3D digitization.

TABLE 1: DESCRIPTION OF LC25000 DATASET

Cancer type	Class name	Class id	Number of samples
Colon Adenocarcinoma	Colon_aca	0	5000
Colon Benign Tissue	Colon_n	1	5000
Lung Adenocarcinoma	Lung_aca	2	5000
Lung Benign Tissue	Lung_n	3	5000
Lung Squamous Cell Carcinoma	Lung_scc	4	5000

V. PERFORMANCE EVALUATION METRICS

In order to build a CNN, many layers are loaded to carry out picture categorization and prediction. A fully linked layer follows components like max-pooling and kernel filters that handle both the training and testing data. All three of the hidden layers utilize the Rectified Linear Unit (ReLU) activation function, with the last layer using the SoftMax function for classification. Metrics including recall, precision, F1-score, and accuracy are used to assess the models' performance. Using these measures, one may choose the best model for the job.

$$Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$Accuracy = TP + TN / (TP + FP + TN + FN)$$

$$F1\ Score = (2 * Recall * Precision) / (Precision + Recall)$$

VI. RESULTS AND DISCUSSION

The accuracy, precision, recall, and F1 score of pre-trained convolutional neural network (CNN) architectures for lung and colon cancer classification are examined in table 2 and figure 3, respectively.

Several pre-trained CNN models were tested on lung and colon cancer classification tasks, and the results show clear variations in performance. EfficientNetB6's performance is below average, with a recall of 0.932, an F1-score of 0.928, and an accuracy of 93%; these are respectable results, but not ideal. A precision, recall, and F1-score of 0.978 and an accuracy of 97% are impressive results for ResNet34, demonstrating its resilience in lung cancer picture classification. The efficiency of VGG-19 in feature extraction and classification tasks is shown by its precision of 0.982, recall of 0.98, F1-score of 0.982, and accuracy of 98%, which further demonstrates its progress.

With outstanding performance, MobileNetV2 outperforms earlier models; it achieves 99% accuracy, F1-score of 0.992, recall of 0.994, and precision of 0.994. With a 99% accuracy rate, F1-score of 0.992 for recall and precision, and a close second for accuracy, ResNet50 demonstrates strong generalizability in the classification challenge. Both MobileNetV2 and ResNet50 demonstrate superior results, making them the most reliable models for this application. These findings confirm that advanced architectures such as MobileNetV2 and ResNet50 provide significant advantages in histopathological image analysis for lung & colon cancer detection.

TABLE 2: EVALUATION OF PRE-TRAINED CNN MODELS FOR LUNG AND COLON CANCER CLASSIFICATION

Model	Precision	Recall	F1 Score	Accuracy
EfficientNetB6	0.928	0.932	0.928	0.93
ResNet34	0.978	0.978	0.978	0.97
VGG-19	0.982	0.98	0.982	0.98
MobileNetV2	0.994	0.994	0.992	0.99
ResNet50	0.992	0.99	0.992	0.99

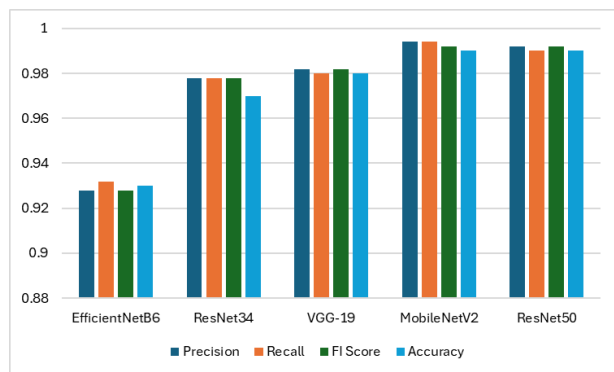


Fig. 3. Performance Evaluating Graph for Pre Trained CNN Architectures in Lung Cancer Classification

VII. CONCLUSION

This work shows that pre-trained CNN architectures may effectively classify histopathology pictures of lung and colon cancers. Thanks to their better feature extraction and generalization capabilities, MobileNetV2 and ResNet50 achieved an astonishing 99% accuracy, making them the most accurate models among those that were evaluated. Notable performance was also shown by ResNet34 and VGG-19, which demonstrated their dependability in cancer detection tasks with accuracies of 97% and 98%, respectively. EfficientNetB6 demonstrated its effectiveness in processing complicated picture data, albeit being little less accurate at 93%. These findings highlight the potential of deep learning, and transfer learning in particular, for precise cancer diagnosis in medical imaging. Pathologists may greatly benefit from the suggested technique, which offers a dependable and effective tool for early cancer detection, ultimately leading to better patient outcomes.

VIII. FUTURE WORK

Future research could focus on enhancing the proposed approach by integrating multimodal data, such as combining histopathological images with genetic profiles and clinical information, to develop a more comprehensive diagnostic system. Additionally, implementing explainable AI techniques, such as Grad-CAM, could make the model's predictions more interpretable and transparent, fostering trust among medical professionals. Expanding the dataset to include diverse samples from multiple institutions would improve the generalizability of the model across various populations and imaging conditions. Real-time application of these models in clinical settings could be explored by optimizing their computational efficiency using lightweight architectures or model compression techniques. Incorporating segmentation-based approaches to focus on specific tumor regions or cellular structures may enhance

classification accuracy further. Future work could also emphasize robustness by training the models to handle noisy or low-quality data more effectively. Comparing CNN models with other advanced architectures, such as Vision Transformers or hybrid deep learning frameworks, could provide insights into alternative solutions. Moreover, extending the model's capabilities to predict patient outcomes, such as prognosis or treatment response, would add valuable clinical relevance. Prospective studies in real-world pathology labs and integration with digital pathology systems, such as whole-slide imaging platforms, could validate and refine the utility of these models in routine diagnostics. Finally, developing automated annotation pipelines would simplify data preparation and enable the efficient handling of large-scale datasets, driving progress in computational pathology.

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REFERENCES

- [1] F. Bray, J. Ferlay, I. Soerjomataram, R. L. Siegel, L. A. Torre, and A. Jemal, "Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA: A Cancer Journal for Clinicians*, vol. 70, no. 4, pp. 313–346, Jul. 2021.
- [2] D. Hanahan and R. A. Weinberg, "Hallmarks of cancer: The next generation," *Cell*, vol. 144, no. 5, pp. 646–674, Mar. 2011.
- [3] A. Bermúdez, I. Arranz-Salas, S. Mercado, J. A. López-Villodres, V. González, F. Rius, et al., "Her2-positive and microsatellite instability status in gastric cancer—clinicopathological implications", *Diagnostics*, vol. 11, no. 6, pp. 944, 2021.
- [4] M. Toğaçar, "Disease type detection in lung and colon cancer images using the complement approach of inefficient sets", *Computers in Biology and Medicine*, vol. 137, pp. 104827, 2021.
- [5] L. F. Sanchez-Peralta, L. Bote-Curiel, A. Picon, F. M. Sanchez-Margallo and J. B. Pagador, "Deep learning to find colorectal polyps in colonoscopy: A systematic literature review", *Artificial intelligence in medicine*, vol. 108, pp. 101923, 2020.
- [6] N. Alimu, A. Maimaiti, M. Maimaitituoxun, H. Muertizha, A. Qukuerhan, Y. Yasheng, J. Yong, N. Mierzhamu, M. Mierzhakemu, A. Aierken et al., "Xrcc1 gene rs72484243 polymorphism is associated with increased laryngeal cancer risk", *Biochemical Genetics*, 2023.
- [7] S. Prabhu, "Pathology of organ systems of the body" in *Textbook of General Pathology for Dental Students*, Springer, pp. 133-146, 2023.
- [8] L. Litjens et al., "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, Dec. 2017.
- [9] H. Janowczyk and A. Madabhushi, "Deep learning for digital pathology image analysis: A comprehensive tutorial with selected use cases," *Journal of Pathology Informatics*, vol. 7, no. 1, p. 29, 2016.
- [10] A. A. Borkowski, M. M. Bui, L. B. Thomas, C. P. Wilson, L. A. DeLand and S. M. Mastorides, "Lung and colon cancer histopathological image dataset (lc25000)", *arXiv preprint*, 2019.
- [11] J. Schmidhuber, "Deep learning in neural networks: An overview," *Neural Networks*, vol. 61, pp. 85–117, Jan. 2015.
- [12] M. Masud, N. Sikder, A.-A. Nahid, A. K. Bairagi and M. A. AlZain, "A machine learning approach to diagnosing lung and colon cancer using a deep learning-based classification framework", *Sensors*, vol. 21, no. 3, pp. 748, 2021.
- [13] N. Kumar, M. Sharma, V. P. Singh, C. Madan and S. Mehandia, "An empirical study of handcrafted and dense feature extraction techniques for lung and colon cancer classification from

- histopathological images", *Biomedical Signal Processing and Control*, vol. 75, pp. 103596, 2022.
- [14] R. R. Wahid, C. Nisa, R. P. Amaliyah and E. Y. Puspaningrum, "Lung and colon cancer detection with convolutional neural networks on histopathological images", *AIP Conference Proceedings*, vol. 2654, 2023.
- [15] A. Krizhevsky, I. Sutskever and G. E. Hinton, "Imagenet classification with deep convolutional neural networks", *Advances in neural information processing systems*, vol. 25, 2012.
- [16] N. Ullah, J. A. Khan, L. A. Alharbi, A. Raza, W. Khan and I. Ahmad, "An efficient approach for crops pests recognition and classification based on novel deeppestnet deep learning model", *IEEE Access*, vol. 10, pp. 73019-73032, 2022.
- [17] N. Ullah, M. S. Khan, J. A. Khan, A. Choi and M. S. Anwar, "A robust end-to-end deep learning-based approach for effective and reliable btd using mr images", *Sensors*, vol. 22, no. 19, pp. 7575, 2022.
- [18] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks", *International conference on machine learning*, pp. 6105-6114, 2019.
- [19] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks", *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4510-4520, 2018.
- [20] M. Pradhan and R. K. Sahu, "Automatic detection of lung cancer using the potential of artificial intelligence (ai)" in *Machine Learning and AI Techniques in Interactive Medical Image Analysis*, IGI Global, pp. 106-123, 2023.
- [21] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, and L. Fei-Fei, "ImageNet large scale visual recognition challenge," *International Journal of Computer Vision*, vol. 115, no. 3, pp. 211–252, 2015.
- [22] P. M. Shakeel, M. A. Burhanuddin, and I. M. Desa, "Automatic lung cancer detection from CT images using improved deep neural network and ensemble classifier," *Neural Computing and Applications*, vol. 34, no. 15, pp. 1–14, 2020.
- [23] A. Rajput and A. Subasi, "Lung cancer detection from histopathological lung tissue images using deep learning" in *Applications of Artificial Intelligence in Medical Imaging*, Elsevier, pp. 51-74, 2023.